

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 3

Formative Assessment (FA-3)

Class: Class 9

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL - Database Management

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which SQL clause is used to combine rows from two or more tables?
 - b. GROUP BY
 - c. JOIN
 - d. UNION
 - e. COMBINE
- b. What is the result of `SELECT COUNT(DISTINCT department) FROM Employees;`?
 - c. Counts all rows
 - d. Counts only distinct departments
 - e. Counts only NULL values
 - f. Error
- c. Which of the following is a DDL command?
 - d. SELECT
 - e. INSERT
 - f. ALTER
 - g. UPDATE
- d. What does the LIKE operator do in SQL?
 - e. Exact match
 - f. Pattern matching
 - g. Range comparison
 - h. NULL check
- e. Which aggregate function ignores NULL values?
 - f. `COUNT(*)`
 - g. `COUNT(column_name)`
 - h. `SUM(column_name)`
 - i. Both B and C

Section II: Short Answer (5 x 2 = 10 marks)

- Write a SQL query to create a table Library with columns: book_id (INT PRIMARY KEY AUTO_INCREMENT), title (VARCHAR 100), author (VARCHAR 50), genre (VARCHAR 30), price (DECIMAL), available (BOOLEAN).
- Differentiate between WHERE and HAVING clauses with two examples each.
- Write SQL queries for a table Students(roll, name, class, section, marks):
- Find students whose name starts with 'A'
- Find students with marks between 50 and 80
- Find students in sections 'A' or 'B'
- Count students in each class
- What is a **Foreign Key**? Explain its role in maintaining referential integrity with an example.
- Write a SQL query to display the second highest salary from an Employees table without using LIMIT or TOP.

Section III: Long Answer (5 x 1 = 5 marks)

- Given the database schema:

```
Students(roll_no, name, class, section)
Subjects(sub_id, sub_name, max_marks)
Marks(roll_no, sub_id, marks_obtained)
```

Write SQL queries to:

- Find the percentage of each student (total marks obtained / total max marks — 100)
- Find the subject with the highest average score
- List students who scored above 90% in any subject
- Find the class topper (highest overall percentage)
- Display a rank list of all students by total marks

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to an SQLite database
- Creates tables School(roll, name, class, section) and Results(roll, sub1, sub2, sub3, sub4, sub5)
- Inserts at least 8 student records with their marks
- Performs and displays:
 - Average marks per student
 - Subject with highest class average
 - Students who need improvement (average below 60%)
 - Grade assignment (A: 90+, B: 75-89, C: 60-74, D: 45-59, F: below 45)

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)